

Demonstration of Proposed Features

Date	8 July
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks
1	Automated Missing Value Imputation	Handles numerical missing values using the mean and categorical values using the mode to clean the dataset automatically.	Implemented	Yes	Successfully handles missing applicant attributes during preprocessing.
2	Categorical Feature Encoding	Converts qualitative applicant metrics (Gender, Education, Property Area) into numerical formats	Implemented	Yes	Integrated cleanly into the pipeline before passing data to the classifiers.

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		optimized for ML training.			
3	Multi-Model Evaluation Pipeline	Trains and evaluates four distinct algorithms: Decision Tree, Random Forest, KNN, and XGBoost.	Implemented	Yes	Achieved 94.7% training and 81.1% testing accuracy with the top-performing XGBoost model.
4	Real-Time Prediction Flask Web App	A interactive user interface built with Flask for users to submit applicant details and instantly view approval results.	Implemented	Yes	Fully functional interface with data entry forms and real-time result rendering.
5	Low-Risk Fast-Track Approval	Identifies stable-income, high-credit applicants to let officers skip manual verification (Scenario 1).	Implemented	Yes	Verified during demo using a mock profile with a good credit history.
6	High-Risk Applicant Detection	Flags applicants with irregular self-employment income and zero credit	Implemented	Yes	Tested successfully; system accurately flagged the profile as high-risk.

S.No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks
		history for manual scrutiny (Scenario 2).			

Feature Implementation Summary:

Metric	Value / Count
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%